MATH 3603–Homework 2

Due: 2-March-2026

1. Let X represent the difference between the number of heads and the number of tails obtained when a coin is tossed n times. (i):What are the possible values of X? (ii): If the coin is assumed fair, then, for $n = 3$, write up the probability mass function of X.

2. Let the probability density of X be given by $f(x) = c(4x - 2x^2)$ for $0 < x < 2$, and 0 otherwise.

(a) What is the value of c?

(b) Compute $\mathbb{P}(1/2 < X < 3/2)$

(c) Compute $\mathbb{E}(X)$ and $\mathrm{Var}(X)$.

3. Let $X_1, X_2, ..., X_n$ be independent random variables with the same distribution function $F(x)$. What is the distribution function of $\max\{X_1, \ldots, X_n\}$ and $\min\{X_1, \ldots, X_n\}$?

4. Suppose that one year has 365 days. There are 100 people in a group, whose birthdays are independent, and each birthday is equally likely to be any day of a year. Let X be the total number of the distinct birthdays in this group. Compute $\mathbb{E}X$ and $\mathrm{Var}(X)$. (Hint: write X as a sum of random variables.)

5. A point (X, Y) is uniformly distributed within the disk of radius 1. That is, its density is

$$f(x,y) = \begin{cases} C, & 0 \le x^2 + y^2 \le 1 \\ 0, & \text{otherwise} \end{cases}$$

(i) What is the value of C? (ii) Compute the marginal probability density of X.